

Chapter 28-Switch MAC Table,LAN,VLAN



1 MAC Address (Media Access Control Address)



What is a MAC Address?

A **MAC Address** is a **unique physical address** assigned to every network device.

-  **Length:** 48 bits (6 bytes)
-  Written in **Hexadecimal format**
-  Permanently stored inside the **Network Interface Card (NIC)**
- Also called **BIA – Burned-In Address**

Example MAC Address:

E8:BA:70:11:28:74

Every device connected to a network has its **own unique MAC address**.



Used for **communication inside a local network (LAN)**.



MAC Address Structure

A MAC address has **two parts**.

Part	Size	Meaning
OUI	First 3 bytes	Identifies manufacturer
Device ID	Last 3 bytes	Unique device number

Example:

E8:BA:70 : 11:28:74

| OUI | | Device ID |

- **OUI** → Assigned to companies (Cisco, Intel, HP, etc.)
- **Device ID** → Unique number for each device produced by that company

Daily Life Example

Think of a **MAC Address** like an Aadhaar Number 

- Every person has **one unique Aadhaar number**
- Every network device has **one unique MAC address**

So inside a LAN, devices recognize each other using **MAC addresses**.

◆ **2 Communication in a Switch Network**

A switch works at **Layer 2 of the OSI model**.

✚ Switch uses **MAC addresses** to send data between devices.

Important point:

✓ Switch **does NOT need IP addresses** to forward frames inside a LAN.

Instead it uses:

MAC Address

3 Switch MAC Address Table

 **What is a MAC Address Table?**

A **MAC Address Table** is a table inside a switch that stores:

- Device **MAC address**
- **Interface/port** where the device is connected

Example table:

MAC Address	Interface
-------------	-----------

AA:BB:CC:11:22:33	Fa0/1
-------------------	-------

DD:EE:FF:44:55:66	Fa0/3
-------------------	-------

This allows the switch to send frames **directly to the correct device**.

How Switch Learns MAC Addresses

Switch learns MAC addresses using:

SOURCE MAC ADDRESS

When a frame arrives:

- 1** Switch checks **source MAC address**
- 2** Switch records it in **MAC table**
- 3** Associates it with the **incoming port**



Example: Communication Between PCs

Imagine a switch connected to **4 PCs**.

PC1

PC2

PC3

PC4

PC1 wants to communicate with **PC3**.

Step 1 – PC1 sends frame

PC1 sends a frame asking:

Who has MAC address of PC3?

Switch does **not know PC3 yet**.



Unknown Unicast Frame

When the switch **doesn't know the destination MAC**, it sends the frame to **all ports except the incoming port**.

This process is called:

UNKNOWN UNICAST FLOODING

Example:

PC1 connected to **Fa0/1**

Switch forwards frame to:

Fa0/2

Fa0/3

Fa0/4

But **not Fa0/1**.



Response from PC3

When **PC3 receives the frame**:

✓ It sends a reply to PC1.

Now the switch learns:

MAC Interface

PC1 Fa0/1

PC3 Fa0/3

Now communication becomes **direct**.



Known Unicast Frame

Now the switch knows where PC3 is.

Next time PC1 sends traffic:

Switch sends frame **ONLY** to Fa0/3

This is called:

KNOWN UNICAST FORWARDING

MAC Address Aging

MAC table entries are **temporary**.

If no traffic comes from a device for **5 minutes**, the switch removes the entry.

MAC Aging Timer = 300 seconds

 Reason:

- Prevents old or invalid entries
 - Keeps MAC table updated
-

Commands to View and Manage MAC Table

1 Show MAC Address Table

show mac address-table

Shows:

- All learned MAC addresses
 - Corresponding switch ports
-

2 Show MAC Table of Specific Interface

show mac address-table interface fa0/1

Shows MAC addresses learned only on **Fa0/1**.

3 Delete All Dynamic MAC Entries

clear mac address-table dynamic

Deletes all **dynamic entries**.

! Static entries remain.

Delete MAC Entries of Specific Interface

```
clear mac address-table dynamic interface fa0/1
```

Deletes entries only for **Fa0/1**.

Delete Specific MAC Address

```
clear mac address-table dynamic address 0001.42ab.23cd
```

Removes one specific MAC entry.



What is a LAN (Local Area Network)?

A **LAN** is a **private network** where devices communicate within the same area.

Examples:



Office network



School network



Home WiFi network

Devices inside a LAN can share:

- Files
 - Printers
 - Internet
 - Applications
-



Broadcast Domain

A **broadcast domain** is a group of devices that receive a **broadcast frame**.

Broadcast MAC address:

FFFF.FFFF.FFFF

Meaning:

Send to ALL devices in the network

⚠️ Problem with Large Broadcast Domains

Too many broadcasts cause:

👤 Performance Issues

Unnecessary traffic reduces network speed.

🔒 Security Issues

Devices can communicate freely without restrictions.

Example:

One employee accessing **confidential HR systems**.

◆ 5 VLAN (Virtual Local Area Network)

📘 What is a VLAN?

A **VLAN** is a logical separation of devices inside a switch.

Even if devices are connected to the same switch, VLANs divide them into **separate networks**.

📌 VLANs work at **Layer 2**.

🧠 Example

Office departments:

Engineering

HR

Sales

Instead of installing **3 physical networks**, we create **3 VLANs**.

VLAN Department

VLAN 10 Engineering

VLAN 20 HR

VLAN 30 Sales

Devices in VLAN 10 **cannot communicate directly** with VLAN 20.

Purpose of VLANs

Improve Network Performance

Broadcast traffic stays **inside the VLAN only**.

This reduces congestion.

Improve Security

Devices from other VLANs cannot access data.

Example:

Engineering PCs cannot access HR systems.

Communication Between VLANs

Switch **cannot route between VLANs**.

For communication between VLANs we need:

Router

This process is called:

INTER-VLAN ROUTING

Methods:

- Router on a Stick
- Layer 3 Switch



Switch Ports



Access Port

An **Access Port** belongs to **only one VLAN**.

Used to connect:

- PCs
- Printers
- Servers

Example:

```
switchport mode access
```



Trunk Port

A **Trunk Port** carries **multiple VLANs**.

Used between:

- Switch ↔ Switch
 - Switch ↔ Router
-



VLAN LAB – Packet Tracer



Network Design

VLAN Name	Network	Gateway
10	Engineering 10.0.0.0/26	10.0.0.62
20	HR 10.0.0.64/26	10.0.0.126
30	Sales 10.0.0.128/26	10.0.0.190



PC IP Configuration

PC	IP	Gateway
PC1	10.0.0.1	10.0.0.62
PC2	10.0.0.2	10.0.0.62
PC3	10.0.0.65	10.0.0.126
PC4	10.0.0.66	10.0.0.126
PC5	10.0.0.129	10.0.0.190
PC6	10.0.0.130	10.0.0.190

Subnet Mask for all PCs:

255.255.255.192



Router Configuration

```
R1> enable
```

```
R1# configure terminal
```

```
interface g0/0
```

```
ip address 10.0.0.62 255.255.255.192
```

```
no shutdown
```

```
interface g0/1  
ip address 10.0.0.126 255.255.255.192  
no shutdown
```

```
interface g0/2  
ip address 10.0.0.190 255.255.255.192  
no shutdown
```

Switch VLAN Configuration

Create VLANs

```
SW1> enable  
SW1# configure terminal
```

```
vlan 10  
name Engineering
```

```
vlan 20  
name HR
```

```
vlan 30  
name Sales
```

Assign Ports to VLANs

```
interface fa3/1  
switchport mode access  
switchport access vlan 10
```

```
interface fa4/1  
switchport mode access  
switchport access vlan 10
```

```
interface fa5/1  
switchport mode access  
switchport access vlan 20
```

```
interface fa6/1
switchport mode access
switchport access vlan 20
```

```
interface fa7/1
switchport mode access
switchport access vlan 30
```

```
interface fa8/1
switchport mode access
switchport access vlan 30
```

Verify VLANs

```
show vlan brief
```

Shows:

- VLAN numbers
- VLAN names
- Ports assigned

Testing Connectivity

Normal Ping

```
PC1 ping 10.0.0.2
```

```
PC1 ping 10.0.0.65
```

Broadcast Ping

Engineering VLAN:

```
PC1 ping 10.0.0.63
```

HR VLAN:

```
PC3 ping 10.0.0.127
```

Sales VLAN:

PC5 ping 10.0.0.191

Packet Tracer Simulation Mode

Steps:

- 1 Select **Simulation Mode**
- 2 Enable **ICMP Filter**
- 3 Send Broadcast Ping
- 4 Click **Capture/Forward**

You will see the **packet flow**.

Expected Result

If Engineering PC sends broadcast:

Device	Receive Broadcast
Engineering PCs	✓ Yes
HR PCs	✗ No
Sales PCs	✗ No

Reason:

Each VLAN = Separate Broadcast Domain

Real Life Example

In a company:

-  Engineering department
-  HR department
-  Sales department

Each department has:

- Separate VLAN
- Separate broadcast traffic
- Better security

But all use **same physical switch infrastructure**.

One-Liners

✓ MAC Address

“A MAC address is a 48-bit unique hardware identifier assigned to a network interface card.”

✓ VLAN

“A VLAN logically separates devices into different broadcast domains on the same physical switch.”

Final Key Points

- ✓ MAC Address = Unique device identifier
 - ✓ Switch learns MAC using **source address**
 - ✓ Unknown unicast = **Flooding**
 - ✓ Known unicast = **Forwarding**
 - ✓ MAC table aging = **5 minutes**
 - ✓ VLAN = Logical network separation
 - ✓ VLAN improves **performance & security**
 - ✓ Each VLAN = **Separate broadcast domain**
-

1 Chapter Summary

◆ MAC Address

- A **MAC Address (Media Access Control Address)** is a **unique hardware identifier** assigned to a **Network Interface Card (NIC)**.
- It is **48 bits long (6 bytes)** and written in **hexadecimal format**.
- Example:
E8:BA:70:11:28:74

Structure:

Part	Size	Purpose
OUI	First 3 bytes	Manufacturer identifier
Device ID	Last 3 bytes	Unique device number

 Used for **communication inside LAN networks**.

◆ Switch MAC Address Table

A **MAC Address Table** is maintained by a switch that maps:

MAC Address Port

Device MAC	Interface where device is connected

The switch **learns MAC addresses dynamically** using the **source MAC address** of incoming frames.

◆ Unknown Unicast Flooding

If the switch **does not know the destination MAC**, it sends the frame to **all ports except the incoming port**.

This is called:

 **Unknown Unicast Flooding**

◆ Known Unicast Forwarding

Once the switch learns the MAC address:

✓ It sends the frame **only to the correct port**.

This improves **network efficiency**.

◆ MAC Address Aging

- Switch removes unused MAC entries after **300 seconds (5 minutes)**.
 - This keeps the MAC table **updated and efficient**.
-

◆ LAN (Local Area Network)

A **LAN** is a network where devices communicate within a **limited geographic area**.

Examples:



Home network



Office network



School network

Devices share:

- Files
 - Printers
 - Internet
 - Applications
-

◆ Broadcast Domain

A **broadcast domain** is a group of devices that receive **broadcast frames**.

Broadcast MAC Address:

FFFF.FFFF.FFFF

Meaning:

 Send packet to **all devices in the network**.

◆ VLAN (Virtual LAN)

A **VLAN** logically separates devices on the same switch into **different networks**.

Example:

VLAN Department

VLAN 10 Engineering

VLAN 20 HR

VLAN 30 Sales

Benefits:

-  Better performance
 -  Better security
 -  Reduced broadcast traffic
-

◆ Inter-VLAN Routing

Devices in different VLANs **cannot communicate directly**.

Communication requires:

- Router
- Layer 3 Switch

This process is called:

 **Inter-VLAN Routing**

◆ Switch Ports

Access Port

Connects end devices.

Example command:

```
switchport mode access  
switchport access vlan 10
```

Trunk Port

Carries traffic from **multiple VLANs**.

Used between:

- Switch ↔ Switch
 - Switch ↔ Router
-

Chapter Conclusion

This chapter explains the **foundation of Layer-2 networking**.

Key ideas:

- ✓ Every device in a network has a **unique MAC address**
- ✓ Switches use **MAC tables** to forward frames efficiently
- ✓ Unknown destinations cause **frame flooding**
- ✓ MAC entries expire using the **aging timer**
- ✓ **LANs** allow devices to communicate locally
- ✓ **Broadcast domains** affect performance in large networks
- ✓ **VLANs logically divide networks** for security and efficiency
- ✓ **Inter-VLAN routing** allows communication between VLANs

 In real enterprise networks, **VLAN segmentation is one of the most important security and performance techniques**.

Q & A

1 What is a MAC Address?

Answer

A **MAC Address (Media Access Control Address)** is a **unique hardware identifier assigned to a network interface card (NIC)**.

Key features:

- 48-bit address
- Written in hexadecimal
- Stored permanently in NIC
- Also called **BIA (Burned-In Address)**

Example:

E8:BA:70:11:28:74

It is used by **Layer 2 devices like switches** to identify devices inside a **Local Area Network (LAN)**.

 Simple explanation:

Think of a MAC address like a **device fingerprint** inside the network.

2 What is the structure of a MAC Address?

A MAC address has **two parts**.

Part	Description
OUI	Organizationally Unique Identifier
Device ID	Unique number for device

Example:

E8:BA:70 : 11:28:74

- First part identifies manufacturer

- Second part identifies device

Companies like **Cisco, Intel, HP** receive OUIs from IEEE.

3 What is a MAC Address Table in a switch?

Answer

A **MAC Address Table** is a table maintained by a switch that stores:

- MAC address
- Associated switch port

Example:

MAC Address	Interface
AA:BB:CC:11:22:33	Fa0/1

This allows the switch to **forward frames efficiently**.

4 How does a switch learn MAC addresses?

Switches learn MAC addresses using the **Source MAC Address** of incoming frames.

Steps:

- 1 Frame arrives at switch
- 2 Switch reads source MAC
- 3 Stores MAC in table
- 4 Associates it with incoming port

This process is called:

 **MAC Learning**

5 What is Unknown Unicast Flooding?

Answer

Unknown unicast flooding occurs when:

- Switch receives a frame
- Destination MAC is **not present in MAC table**

Switch sends the frame to **all ports except the incoming port**.

Purpose:

- ✓ Find the destination device.

6 What is Known Unicast Forwarding?

Answer

When the switch **knows the destination MAC address**, it forwards the frame only to the correct port.

Benefits:

- ✓ Faster communication
- ✓ Less network traffic

7 What is MAC Address Aging?

Answer

MAC table entries are **temporary**.

If a device does not send traffic for **300 seconds (5 minutes)**, the switch removes the entry.

Purpose:

- ✓ Prevent outdated entries
 - ✓ Maintain table efficiency
-

8 What is a LAN?

A **Local Area Network (LAN)** is a network where devices communicate within a **limited geographic area**.

Examples:

-  Home network
-  Office network
-  School network

LANs typically use:

- Ethernet
- WiFi

9 What is a Broadcast Domain?

A **broadcast domain** is a network segment where broadcast packets reach **all devices**.

Broadcast address:

FFFF.FFFF.FFFF

Problems with large broadcast domains:

-  Network congestion
-  Security risks

10 What is a VLAN?

A **VLAN (Virtual Local Area Network)** logically separates devices on a switch into **different broadcast domains**.

Example:

VLAN Department

10 Engineering

20 HR

VLAN Department

30 Sales

Benefits:

- ✓ Security
- ✓ Traffic isolation
- ✓ Better performance

1 1 Why VLANs are used?

Main reasons:

-  Reduce broadcast traffic
-  Improve security
-  Improve performance
-  Organize departments

1 2 Can devices in different VLANs communicate?

No.

Devices in different VLANs require:

Inter-VLAN Routing

This can be done using:

- Router on a Stick
- Layer 3 Switch

1 3 What is an Access Port?

An **Access Port** belongs to a single VLAN.

Used for:

- PCs
- Printers

- Servers

Example command:

```
switchport mode access  
switchport access vlan 10
```

1 4 What is a Trunk Port?

A **Trunk Port** carries traffic for multiple VLANs.

Used between:

- Switch ↔ Switch
- Switch ↔ Router

It uses **VLAN tagging**.

1 5 What command shows MAC table?

```
show mac address-table
```

Displays:

- Learned MAC addresses
 - Associated ports
 - VLAN information
-

1 6 How do you clear MAC table?

Delete all entries:

```
clear mac address-table dynamic
```

Delete specific interface entries:

```
clear mac address-table dynamic interface fa0/1
```

Delete specific MAC:

```
clear mac address-table dynamic address 0001.42ab.23cd
```

1 7 What command verifies VLAN configuration?

show vlan brief

Displays:

- VLAN ID
 - VLAN name
 - Assigned ports
-

1 8 Real-World Interview Scenario Question

Question

A user from HR cannot communicate with Engineering PC. Why?

Answer

Because:

- HR PC is in **VLAN 20**
- Engineering PC is in **VLAN 10**

Different VLANs cannot communicate without **Inter-VLAN routing**.

Solution:

Configure:

- Router
 - Layer 3 Switch
-

1 9 What happens when a broadcast packet is sent?

Broadcast packets are delivered to:

✓ All devices in same VLAN

Devices in other VLANs:

✗ Do not receive it.

2 0 What are advantages of VLAN?

Major benefits:

- ✓ Reduced broadcast traffic
 - ✓ Improved security
 - ✓ Better network organization
 - ✓ Logical segmentation
-

★ “Explain VLAN in one line.”

Answer:

“A VLAN logically separates devices into different broadcast domains on the same physical switch to improve network security and performance.”

Final Important Points to Remember

- ✓ MAC Address = Unique device identifier
 - ✓ Switch learns MAC from **source address**
 - ✓ Unknown unicast causes **flooding**
 - ✓ Known unicast allows **direct forwarding**
 - ✓ MAC aging timer = **300 seconds**
 - ✓ VLAN divides network into **multiple broadcast domains**
 - ✓ Inter-VLAN routing allows VLAN communication
-